

TELECOM CHURN ANALYSIS: FINAL STRATEGIC RECOMMENDATIONS & ACTION PLAN

1. Executive Summary & Financial Imperative

An end-to-end data pipeline was constructed to combat customer churn within a 10,000-user database. By aggregating raw telecom variables (call duration, complaints, recharge frequency) via MySQL and deploying a Random Forest binary classification model, we successfully isolated the exact drivers of churn. Crucially, SHAP (Shapley Additive exPlanations) provided granular model explainability, moving the analysis from reactive metrics to prescriptive retention strategies. High churn directly inflates Customer Acquisition Costs (CAC); deploying the strategies below will directly protect top-line revenue.

2. The Threat of the 'Frustration Index'

Our engineered SQL metric, the *Frustration Index* (combining network drops and unresolved complaints weighted mathematically), was flagged by the SHAP algorithm as a premier driver of churn. Customers are not leaving due to price; they are leaving due to unresolved technical friction.

- **Operational Recommendation:** Overhaul the IVR (Interactive Voice Response) routing logic. If a customer's Frustration Index crosses a threshold of 8 (identified via SQL querying), their next call must instantly bypass standard Tier-1 support. Route them directly to a specialized "Retention Resolution" team empowered with the authority to grant immediate billing credits and tech-dispatch priority.

3. Contract Volatility & Incentive Restructuring

SHAP visualizations confirm that Month-to-Month contracts exhibit the highest risk density. The lack of an exit barrier creates immense volatility.

- **Marketing Recommendation:** Cease offering flat discounts on Month-to-Month renewals. Instead, reallocate that marketing capital to aggressively subsidize 1-Year or 2-Year contract

upgrades. Offer tangible lifecycle benefits—such as a free 6-month streaming service subscription or a free router upgrade—exclusively for converting to an annual contract.

4. The 5G Upgrade Imperative (ROI of Infrastructure)

The machine learning model isolated network architecture as a major retention anchor. Users on the 5G network display a drastically lower baseline churn probability than legacy 4G users.

- **Sales Recommendation:** Launch aggressive, localized 5G upgrade campaigns targeting our "At Risk" 4G segment. The data suggests that subsidizing the cost of a 5G-compatible SIM card or device upgrade for an "At Risk" user is mathematically cheaper than the marketing cost required to acquire a net-new customer to replace them.

5. Segment-Specific Campaign Execution The database has been scored by the AI and segmented into three actionable tiers: Loyal, At Risk, and Dormant. Standardized "blast" marketing is inefficient; we must deploy targeted actions:

- **Dormant Segment (High days since recharge, near-zero usage):** These users have practically churned but haven't formally cancelled. Deploy automated "Win-Back" SMS campaigns offering heavy 48-hour data-bonus incentives to trigger a baseline recharge and reactivate their habit loop.
- **At Risk Segment (High Frustration Index or High AI Churn Probability):** Deploy proactive, outbound outreach. Do not wait for them to call to complain. Issue preemptive apologies via email/SMS alongside loyalty data-booster for known, localized network drop events.
- **Loyal Segment (Low Churn Probability):** Suppress discount-based retention marketing to this group to protect gross margins. Instead, engage them with Referral Programs, turning our most stable users into acquisition channels.